Trace Google Gemini Models in Langfuse

This notebook shows how to trace and observe Google Gemini models with Langfuse and the Google GenAI SDK.

> **What is Google Gemini?** Google Gemini is Google's family of multimodal generative models (text, images, audio, video, code) available through the Gemini API and Vertex AI, with tiers like Flash and Pro for different speed/quality needs.

> **What is the Google GenAI SDK?** The Google GenAI SDK is a unified client library (Python/JavaScript) that simplifies calling Gemini—handling auth (API key or ADC), streaming, tool/function calling, and safety—so you can integrate models in a few lines.

> **What is Langfuse?** Langfuse is an open source platform for LLM observability and monitoring. It helps you trace and monitor your AI applications by capturing metadata, prompt details, token usage, latency, and more.

Step 1: Install Dependencies

Before you begin, install the necessary packages in your Python environment:

```
%pip install google-genai openai langfuse openinference-instrumentation-goog
```

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Step 2: Configure Langfuse SDK

Next, set up your Langfuse API keys. You can get these keys by signing up for a free Langfuse Cloud account or by self-hosting Langfuse. These environment variables are essential for the Langfuse client to authenticate and send data to your Langfuse project.

Also set your Google Vertex API credentials which uses Application Default Credentials (ADC) from a service account key file.

```
import os

# Get keys for your project from the project settings page
# https://cloud.langfuse.com
LANGFUSE_SECRET_KEY ="sk-lf-afada7dc-2d00-4be8-b7d5-8759a31db192"
LANGFUSE_PUBLIC_KEY ="pk-lf-c1267e7e-1106-4389-ab27-da266102c1fa"
```

```
LANGFUSE_BASE_URL = "https://cloud.langfuse.com" # EU EU region
# LANGFUSE_BASE_URL = "https://us.cloud.langfuse.com" # US US region

# Your openai key
# We use OpenAI for this demo, could easily change to other models
os.environ["GEMINI_API_KEY"] ="AIzaSyDw18J5Ze-PxABaHV_ljdfl8Da8vBgr018"
```

With the environment variables set, we can now initialize the Langfuse client. `get_client()` initializes the Langfuse client using the credentials provided in the environment variables.

```
from langfuse import get_client

# Initialise Langfuse client and verify connectivity
langfuse = get_client()
```

```
WARNING:langfuse:Authentication error: Langfuse client initialized without pu
```

Step 3: OpenTelemetry Instrumentation

Use the `GoogleGenAIInstrumentor` library to wrap Google GenAI SDK calls and send OpenTelemetry spans to Langfuse.

```
from openinference.instrumentation.google_genai import GoogleGenAIInstrument

GoogleGenAIInstrumentor().instrument()
```

Step 4: Run an Example

```
from google import genai

client = genai.Client()

response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents="What is Langfuse?",
)
print(response.text)
```

```
**Langfuse is an open-source LLM observability and evaluation platform.**

In simpler terms, it's a tool that helps developers monitor, debug, and impro

Think of it as the "black box recorder" or "Google Analytics" specifically de

Here's a breakdown of what Langfuse does and why it's important:
```

```
1.  **LLM Observability & Monitoring:**
    *   **Traces:** Langfuse captures detailed "traces" of every interaction
    *   **Visual Debugging:** It provides a user interface to visualize these
    *   **Metrics & Analytics:** It aggregates data on usage, costs, latency,

2.  **Debugging & Troubleshooting:**
    *   When an LLM application behaves unexpectedly (e.g., gives a nonsensic
    *   It helps pinpoint issues in prompts, RAG logic, tool definitions, or

3.  **Evaluation & Improvement:**
    *   **Human Feedback:** You can collect human feedback (e.g., "good answe
    *   **Automated Evaluation:** It supports running automated evaluations (
    *   **Experimentation:** Langfuse facilitates A/B testing and comparing d

**Key Features & Components:**

*   **SDKs:** Provides client libraries (Python, JS/TS) to easily integrate i
*   **Dashboard & UI:** A web interface to view traces, manage projects, anal
*   **Prompt Management:** Helps version and manage your prompts.
*   **Open-Source:** The core platform is open-source, allowing for self-host

**Why is it important?**

Developing robust LLM applications is challenging due to the non-deterministi
*   Understand complex LLM behaviors.
*   Debug issues quickly and efficiently.
*   Iterate on prompts and application logic with confidence.
*   Ensure the quality and reliability of LLM applications in production.
*   Manage costs and performance.

In essence, Langfuse bridges the gap between developing LLM-powered features
```

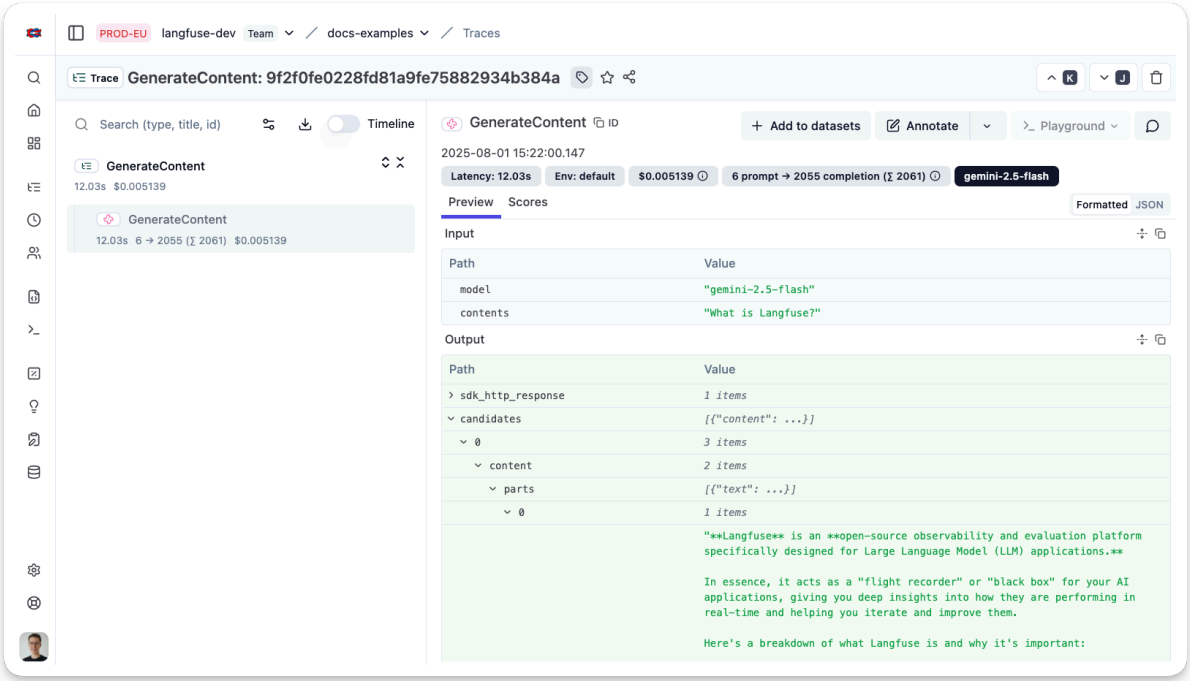
```
# Streaming Example
for chunk in client.models.generate_content_stream(
    model="gemini-2.5-flash",
    contents="What is Langfuse?",
):
    print(chunk.text, end="", flush=True)
print()  # newline after streaming
```

```
**Langfuse** is an **open-source observability and evaluation platform for La

In simpler terms, it's a tool that helps developers building applications pow

1.  **Understand what's happening under the hood (Observability):**
    *   **Trace and Log Everything:** It records every step of an LLM call or
    *   **Visualize Flows:** It presents these traces in an intuitive waterfa
    *   **Debug Issues:** When an LLM application doesn't behave as expected
    *   **Monitor Performance:** Track metrics like latency, token usage, cos

2.  **Improve the quality and reliability of their applications (Evaluation):
    *   **Human Annotation:** Allows developers or human testers to provide f
    *   **Automated Evaluation:** Enables the use of LLMs themselves (LLM-as-
    *   **Experimentation:** Facilitates A/B testing different prompts, model
    *   **Dataset Management:** Helps in building and managing datasets for t

**Why is Langfuse needed?**
```

```
LLM applications are inherently non-deterministic and can be complex. Traditi
*   Make the black box of LLM interactions more transparent.
*   Systematically test and iterate on prompt engineering and application log
*   Ensure the application performs reliably and cost-effectively in producti

**Key Features/Benefits at a Glance:**

*   **End-to-end Tracing:** From user input to final output, including all in
*   **Detailed Logging:** Captures prompts, responses, metadata, and more.
*   **Cost & Latency Monitoring:** Keep an eye on operational expenses and pe
*   **Human & Automated Evaluation:** Tools to measure and improve output qua
*   **Experimentation & A/B Testing:** Compare different configurations.
*   **Open Source:** Provides flexibility for self-hosting and community cont
*   **SDKs & Integrations:** Works with popular frameworks like LangChain and

In essence, Langfuse acts like a "flight recorder" and "quality assurance lab
```

View Traces in Langfuse

After executing the application, navigate to your Langfuse Trace Table. You will find detailed traces of the application's execution, providing insights into the agent conversations, LLM calls, inputs, outputs, and performance metrics.



View trace in Langfuse

https://cloud.langfuse.com/project/cloramnkj0002jz088vzn1ja4/traces/9f2f0fe0228fd81a9fe75882934b384a?timestamp=2025-08-01T13%3A22%3A00.147Z&display=details&observation=b7a63ca7e1d083bc